

International Day for the Preservation of the Ozone Layer

SENIORDTECH

UNITReliever
Classwork**COMPONENT**

Technology

THEMEInternational
Day for the
Preservation of
the Ozone
Layer**DATE**

2026-09-16

Aim(s) of Lesson

Students will explain the relationship between technological development, ozone-depleting substances, and environmental consequences; evaluate digital sources; analyse evidence, regulation, and redesign; and produce a concise evidence-based digital brief.

Resources Required

Chromebook with internet access; Google Slides or Google Docs; UN Ozone Day and Montreal Protocol information; NZ Ministry for the Environment information; normal school submission/sharing system.

Preparation

Confirm students can access Google Slides or Google Docs. Display the required outcome and source-check questions. Students should work independently, paraphrase evidence, distinguish fact from interpretation, and record sources. The outcome is one-page or one-slide, not a long presentation.

Health & Safety

Students remain seated while using Chromebooks, keep bags and charging leads clear of walkways, use school-approved websites, avoid entering personal information into unfamiliar sites, and follow normal expectations for appropriate internet use. Students should not rely on AI-generated summaries as evidence unless the original source is checked. No workshop hazards are involved.

Starter

5 mins. Prompt: When a technology produces major benefits but later causes environmental harm, what evidence should be required before society changes or restricts it? Students write 3-4 sentences. Distinguish ozone depletion from climate change and frame the lesson as a technological-systems case study.

Demonstration

10 mins. Introduce the systems chain: NEED -> TECHNOLOGICAL DEVELOPMENT -> WIDESPREAD USE -> UNINTENDED CONSEQUENCE -> SCIENTIFIC EVIDENCE -> POLICY/REGULATION -> TECHNOLOGICAL REDESIGN -> OUTCOME. Explain CFCs, ozone loss, the Montreal Protocol, and source evaluation using Authority, Evidence, Currency, and Corroboration. Model a concise evidence brief for a school audience.

Practice

40 mins. Research using at least three credible sources, including an authoritative international or NZ government/scientific source. Create one-page evidence brief or one-slide visual brief titled Ozone Layer: A Technology Systems Case Study. Include context, evidence, response, impact, an Aotearoa NZ connection, a justified 80-120 word technology judgement, clear hierarchy, concise writing, meaningful visualisation, and acknowledged sources. Y11 explains and justifies; Y12 analyses relationships and trade-offs; Y13 critically evaluates limitations, competing considerations, and future decision-making. Verify one claim against a second independent source.

Plenary

5 mins. Pair exchange: identify the strongest evidence in a partner's brief and one claim that could be better supported or qualified. Exit statement: The most important lesson this case gives future technologists is... Save as Surname_OzoneDay_2027 and submit as normal.

Homework

None. If unfinished, students may complete the evidence brief if normal class procedures allow.

Lesson Evaluation

What went well? What did not go well? What changes are needed? How will impact be gathered? Did changes from last time make a difference? What are the next steps? Other comments: This is a stand-alone lesson. Senior progression is based on depth of thinking rather than separate tasks.